

ASSIGNMENT-1

A linked list is a dynamic linear data structure where elements are stored in separate memory locations and connected using pointers. Each element, called a node, contains data and a pointer to the next node in the sequence. Unlike arrays, linked lists do not require contiguous memory allocation, making insertion and deletion operations more efficient. Common types include singly linked lists, doubly linked lists, and circular linked lists. Linked lists are widely used in implementing stacks, queues, hash tables, and memory management systems. Their flexibility allows efficient handling of data that changes frequently during program execution.